Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	3 (Term 1)	**Year:**	8
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to explain how data is split into packets and model forwarding between networks. They will also learn to simulate packet delivery and explain how ordering and retry can support a complete message. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Check a correct role for routers and servers and an explanation of why a message can be sent in smaller parts. Check the reconstructed message and an explanation that recovery depends on the communication protocol, not a guarantee from every router. Note pupils who need further guided practice.

LESSON SLOTS - MONDAY AND TUESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	7 September 2026	By the end of this lesson, students will be able to explain how data is split into packets and model forwarding between networks.	**H:** Ask whether a large image needs to travel across the internet as one indivisible object. **ID:** Explain packets, destination network addresses and routers forwarding them; servers provide services rather than acting as every intermediate router. **WD:** Split a short message into numbered classroom cards and trace two possible paths through router nodes; clarify that real routing follows network rules, not coin tosses. **YD:** Pupils build a small sender-router-receiver diagram and explain the destination information needed to forward a card. **C:** Check a correct role for routers and servers and an explanation of why a message can be sent in smaller parts.	**Support:** Provide labelled hardware cards, a simple network map and a check-observe-report flow. Model one path at a time and accept spoken reasoning before writing. **Extension:** Ask pupils to compare two possible paths or solutions and explain one limit of the network model or chosen storage arrangement.

2	8 September 2026	By the end of this lesson, students will be able to simulate packet delivery and explain how ordering and retry can support a complete message.	**H:** Give a group message cards in a different order and ask how to reconstruct the text. **ID:** Explain that the model illustrates packet forwarding and TCP-style ordering and retry; IP alone does not guarantee arrival or order. **WD:** Send numbered cards along agreed routes, delay one card and model the receiver identifying the missing part and the sender resending it. **YD:** Groups run the simulation, record arrival order, reconstruct the message and explain a missing-card scenario. **C:** Check the reconstructed message and an explanation that recovery depends on the communication protocol, not a guarantee from every router.	**Support:** Provide labelled hardware cards, a simple network map and a check-observe-report flow. Model one path at a time and accept spoken reasoning before writing. **Extension:** Ask pupils to compare two possible paths or solutions and explain one limit of the network model or chosen storage arrangement.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Lower Secondary Computing Student Book 8 printed pages 14-21
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Labelled network diagrams, hardware cards, message cards and connection scenarios
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Teacher reference: https://www.rfc-editor.org/rfc/rfc9293.html (TCP and reliable delivery)



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?